

A Soft Introduction to Covariance Matrix Adaptation Evolution Strategy (CMA-ES)

Optimization Algorithm Team

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Introduction

Optimization with Gradient Descents

Random Search

Evolution Strategy: (μ, λ) -ES

Ill-Conditioned Problems

Covariance Matrix Adaptation Evolution Strategy (CMA-ES)

Introduction

The objective of the study is to understand the progression

gradient descent → random search → evolution strategy → CMA-ES

After: we shall see how important it is to adapt a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

Modified Objective: **Visit CMA-ES**

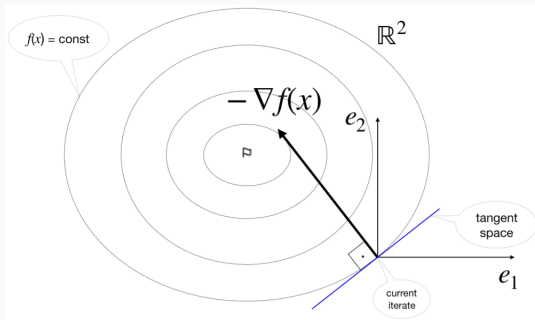
Optimization with Gradient Descents

Suppose we want to solve

$$\min_{x \in D} f(x)$$

If f is **smooth**, we use
gradient information

$$x_{n+1} = x_n - \sigma \nabla f(x_n)$$



Search information

GD uses **gradient information** to search for the optimizer.

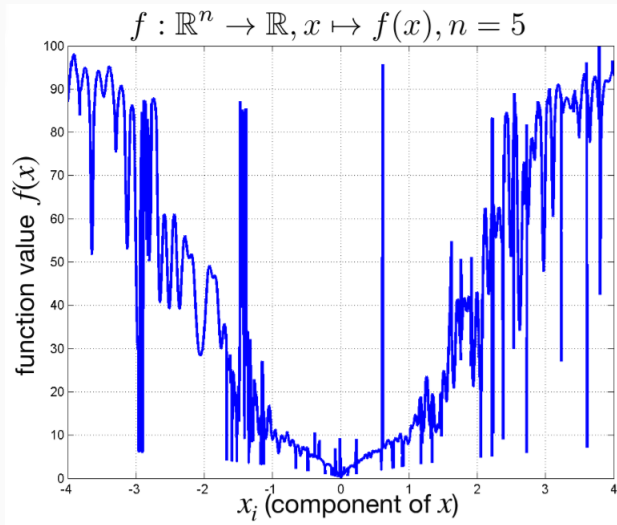


Figure 1: 5D Rugged Landscape ¹

¹Hansen [2019]

Random Search

How can we reach the optimizer?

Random Search ²

With a starting point $m \Rightarrow$ generate random points around $m \Rightarrow$ calculate the function values f .

Search information

The **random search** uses a point m to generate **random points** based on probability distribution and calculate **the fitness value (function value)**.

Drawback: No directional learning/adaptation.

This is as stated " Finding a needle in an n-dimensional haystack " ³.

²White [1970]

³Korn [1966]

Evolution Strategy: (μ, λ) -ES

A simulation is here!

Search Information

(μ, λ) -ES uses a mean m generate to λ random points (called *offsprings*) and selects μ points (called *parents*), recombine to generate a new mean, and repeat the process.

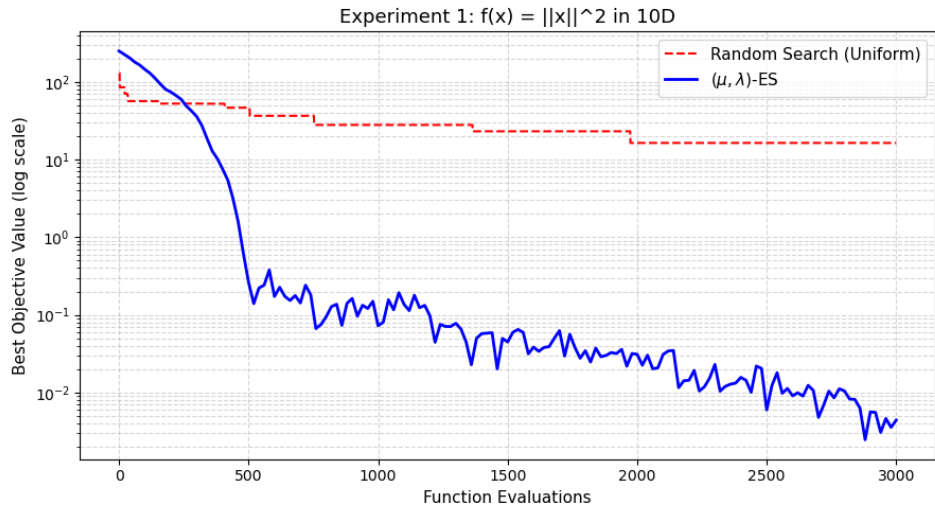


Figure 2: Comparison between random search with uniform distribution and (μ, λ) -ES.

III-Conditioned Problems

(μ, λ) -ES is not enough...

- One observation is that the algorithm is inefficient when the successful search directions from a thin elongated ellipsoid.

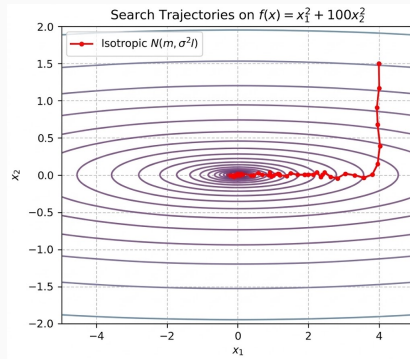
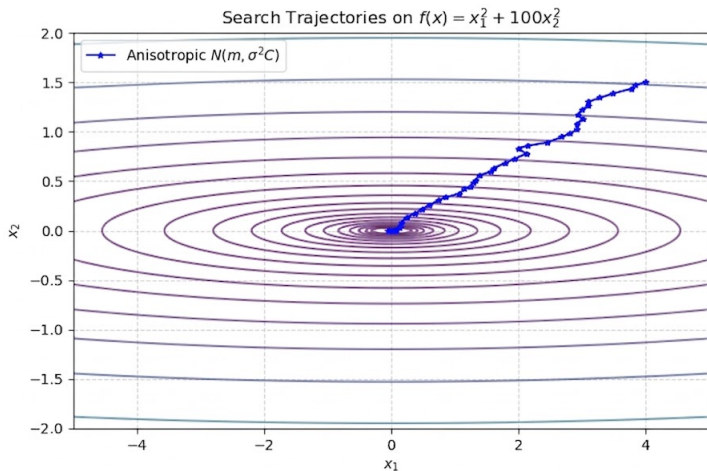


Figure 3: Function f has anisotropic level set. (Steep valey in x_2 and flat in x_1).

Preferable search trajectory



Drawback:

Have a direction, but ineffective with ill-conditioned problems.

Covariance Matrix Adaptation Evolution Strategy (CMA-ES)

Search Information

CMA-ES = (μ, λ) -ES + learning/adaptation

(μ, λ) -ES generates points

$$X \sim \mathcal{N}(m, \sigma^2 I).$$

CMA-ES generates points

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

(μ, λ) -ES generates points

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CMA-ES generates points

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

What does the covariance matrix C do here?

A simulation is here!

- Directions in which successful steps have repeatedly occurred should receive larger variance in the future generations.

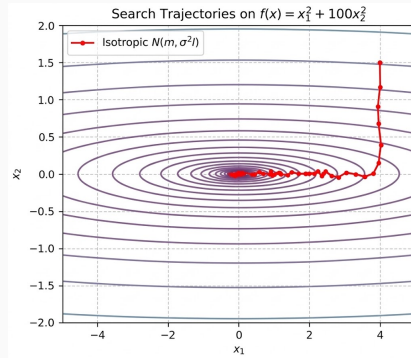


Figure 4: Function f has anisotropic level set. (Steep valey in x_2 and flat in x_1).

Search Information

CMA-ES learns not only where to search though the starting point m but also how to search through C .

What information that C gives?⁴

$$C_{n+1} = (1 - c_\mu) C_n + c_\mu \frac{1}{\sigma(i)^2} C_{n+1}^\mu$$

⁴Aguirre et al. [2026]

What information that C gives?⁴

$$C_{n+1} = \left(\underbrace{1 - c_\mu}_{\text{learning rate}} \right) C_n + c_\mu \frac{1}{\sigma(i)^2} \underbrace{C_{n+1}^\mu}_{\text{reproduce the best candidates}}$$

⁴Aguirre et al. [2026]

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⁴Aguirre et al. [2026]

There's something more...

We can control the step size σ^4

$$\ln \sigma_{n+1} = \ln \sigma_n + \frac{c_\sigma}{d_\sigma} \left(\frac{\|p_{n+1}^\sigma\|}{E \|\mathcal{N}(0, I)\|} - 1 \right)$$

⁴Aguirre et al. [2026]

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⁴Aguirre et al. [2026]

The End of CMA-ES!!!

Gradient-based Opt.

Gradient Descent

Derivative-Free Opt.
Random Search

Evolution Strategy

CMA-ES

References

- Nikolaus Hansen. How to evolve gradient descent into evolution strategies and CMA-ES. Presentation slides, Symposium on Evolutionary Algorithms: Back to the Future and Beyond, Delft, Netherlands, September 2019. URL <http://www.cmap.polytechnique.fr/~nikolaus.hansen/cmaes-from-gradient-descent-delft2019.pdf>.
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**Thank you for your attentions!!!
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